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SPRING REST USING SPRING BOOT

1)Create a Spring Web Project using Maven

SpringLearnApplication.java

```
package com.cognizant.springlearn;
```

```
import org.slf4j.Logger;
```

```
import org.slf4j.LoggerFactory;
```

```
import org.springframework.boot.SpringApplication;
```

```
import org.springframework.boot.autoconfigure.SpringBootApplication;
```

```
@SpringBootApplication
```

```
public class SpringLearnApplication {
```

```
    private static final Logger LOGGER =
```

```
        LoggerFactory.getLogger(SpringLearnApplication.class);
```

```
    public static void main(String[] args) {
```

```
        LOGGER.info("START");
```

```
        SpringApplication.run(SpringLearnApplication.class, args);
```

```
        LOGGER.debug("Application is running successfully.");
```

```
        LOGGER.info("END");
```

```
    }
```

```
}
```

SpringLearnApplicationTests.java

```
package com.cognizant.springlearn;

import org.junit.jupiter.api.Test;
import org.springframework.boot.test.context.SpringBootTest;

@SpringBootTest
class SpringLearnApplicationTests {

    @Test
    void contextLoads() {
    }

}
```

application.properties

```
logging.level.org.springframework=info
logging.level.com.cognizant=debug
spring.application.name=spring-learn
server.port=8080
```

pom.xml

```
<dependencies>

    <!-- Spring Web -->

    <dependency>

        <groupId>org.springframework.boot</groupId>

        <artifactId>spring-boot-starter-web</artifactId>

    </dependency>

    <!-- Spring Boot DevTools -->

    <dependency>

        <groupId>org.springframework.boot</groupId>
```

```
<artifactId>spring-boot-devtools</artifactId>
<scope>runtime</scope>
<optional>true</optional>
</dependency>
```

```
<!-- Testing -->
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-test</artifactId>
  <scope>test</scope>
</dependency>
```

```
</dependencies>
```

Spring Core – Load Country from Spring Configuration XML

country.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="
    http://www.springframework.org/schema/beans
    https://www.springframework.org/schema/beans/spring-beans.xsd">

  <bean id="country" class="com.cognizant.springlearn.Country">

    <property name="code" value="IN"/>
```

```
<property name="name" value="India"/>
```

```
</bean>
```

```
</beans>
```

Country.java

```
package com.cognizant.springlearn;
```

```
public class Country {
```

```
    private String code;
```

```
    private String name;
```

```
    public Country() {
```

```
        System.out.println("Inside Country Constructor.");
```

```
    }
```

```
    public String getCode() {
```

```
        return code;
```

```
    }
```

```
    public void setCode(String code) {
```

```
        System.out.println("Inside setCode()");
```

```
        this.code = code;
```

```
    }
```

```
    public String getName() {
```

```
        return name;
```

```
    }
```

```
    public void setName(String name) {
```

```
        System.out.println("Inside setName()");
```

```
        this.name = name;
```

```
    }
```

```
    @Override
```

```
    public String toString() {
```

```
        return "Country [code=" + code + ", name=" + name + "];"
```

```
}  
}
```

SpringLearnApplication.java

```
package com.cognizant.springlearn;  
  
import org.slf4j.Logger;  
import org.slf4j.LoggerFactory;  
import org.springframework.context.ApplicationContext;  
import org.springframework.context.support.ClassPathXmlApplicationContext;  
  
public class SpringLearnApplication {  
  
    private static final Logger LOGGER =  
        LoggerFactory.getLogger(SpringLearnApplication.class);  
  
    public static void displayCountry() {  
  
        ApplicationContext context =  
            new ClassPathXmlApplicationContext("country.xml");  
  
        Country country = context.getBean("country", Country.class);  
  
        LOGGER.debug("Country : {}", country);  
    }  
  
    public static void main(String[] args) {  
  
        displayCountry();  
  
    }  
}
```

pom.xml

```
<dependency>  
    <groupId>org.springframework</groupId>  
    <artifactId>spring-context</artifactId>  
</dependency>
```

application.properties

```
server.port=8083
```

```
spring.application.name=spring-learn
```

Hello World RESTful Web Service

HelloController.java

```
package com.cognizant.springlearn.controller;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
public class HelloController {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(HelloController.class);

    @GetMapping("/hello")
    public String sayHello() {

        LOGGER.info("START");

        String message = "Hello World!!";

        LOGGER.info("END");

        return message;
    }
}
```

SpringLearnApplication.java

```
package com.cognizant.springlearn;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class SpringLearnApplication {
```

```
    public static void main(String[] args) {  
        SpringApplication.run(SpringLearnApplication.class, args);  
    }  
  
}
```

application.properties

```
server.port=8083  
spring.application.name=spring-learn
```

pom.xml Dependency

```
<dependency>  
    <groupId>org.springframework.boot</groupId>  
    <artifactId>spring-boot-starter-web</artifactId>  
</dependency>
```

REST - Country Web Service

Country.java

```
package com.cognizant.springlearn;  
  
public class Country {  
  
    private String code;  
    private String name;  
  
    public Country() {  
    }  
  
    public String getCode() {  
        return code;  
    }  
  
    public void setCode(String code) {  
        this.code = code;  
    }  
  
    public String getName() {  
        return name;  
    }  
}
```

```

    public void setName(String name) {
        this.name = name;
    }
}

```

country.xml

```

<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
           https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="country"
          class="com.cognizant.springlearn.Country">

        <property name="code" value="IN"/>
        <property name="name" value="India"/>

    </bean>

</beans>

```

CountryController.java

```

package com.cognizant.springlearn.controller;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;

import com.cognizant.springlearn.Country;

@RestController
public class CountryController {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(CountryController.class);

    @RequestMapping("/country")

```



```

public Country getCountryIndia() {

    LOGGER.info("START");

    ApplicationContext context =
        new ClassPathXmlApplicationContext("country.xml");

    Country country = context.getBean("country", Country.class);

    LOGGER.info("END");

    return country;
}
}

```

SpringLearnApplication.java

```

package com.cognizant.springlearn;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class SpringLearnApplication {

    public static void main(String[] args) {
        SpringApplication.run(SpringLearnApplication.class, args);
    }

}

```

REST - Get country based on country code

Country.java

```

package com.cognizant.springlearn.model;

public class Country {

    private String code;
    private String name;

    public Country() {

```

```

    }

    public Country(String code, String name) {
        this.code = code;
        this.name = name;
    }

    public String getCode() {
        return code;
    }

    public void setCode(String code) {
        this.code = code;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }
}

```

country.xml

```

<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
        http://www.springframework.org/schema/beans
        https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="countryList" class="java.util.ArrayList">
        <constructor-arg>
            <list>

                <bean class="com.cognizant.springlearn.model.Country">
                    <property name="code" value="US"/>
                    <property name="name" value="United States"/>
                </bean>

                <bean class="com.cognizant.springlearn.model.Country">

```

```

        <property name="code" value="DE"/>
        <property name="name" value="Germany"/>
    </bean>

    <bean class="com.cognizant.springlearn.model.Country">
        <property name="code" value="IN"/>
        <property name="name" value="India"/>
    </bean>

    <bean class="com.cognizant.springlearn.model.Country">
        <property name="code" value="JP"/>
        <property name="name" value="Japan"/>
    </bean>

</list>
</constructor-arg>
</bean>

```

```
</beans>
```

CountryService.java

```

package com.cognizant.springlearn.service;

import java.util.List;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.stereotype.Service;

import com.cognizant.springlearn.model.Country;

@Service
public class CountryService {

    public Country getCountry(String code) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext("country.xml");

        List<Country> countryList =
            (List<Country>) context.getBean("countryList");

        for (Country country : countryList) {

```

```

        if (country.getCode().equalsIgnoreCase(code)) {
            return country;
        }
    }

    return null;
}
}

```

CountryController.java

```

package com.cognizant.springlearn.controller;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;

import com.cognizant.springlearn.model.Country;
import com.cognizant.springlearn.service.CountryService;

@RestController
public class CountryController {

    @Autowired
    private CountryService countryService;

    @GetMapping("/countries/{code}")
    public Country getCountry(@PathVariable String code) {

        return countryService.getCountry(code);

    }

}

```

SpringLearnApplication.java

```

package com.cognizant.springlearn;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication

```

```
public class SpringLearnApplication {  
  
    public static void main(String[] args) {  
  
        SpringApplication.run(SpringLearnApplication.class, args);  
  
    }  
  
}
```

Create authentication service that returns JWT

pom.xml

```
<dependencies>  
    <dependency>  
        <groupId>org.springframework.boot</groupId>  
        <artifactId>spring-boot-starter-security</artifactId>  
    </dependency>  
</dependencies>
```

AuthenticationResponse.java

```
package com.cognizant.springlearn.model;  
  
public class AuthenticationResponse {  
  
    private String token;  
  
    public AuthenticationResponse() {  
    }  
  
    public AuthenticationResponse(String token) {  
        this.token = token;  
    }  
  
    public String getToken() {  
        return token;  
    }  
}
```

```
    public void setToken(String token) {  
        this.token = token;  
    }  
}
```

JwtUtil.java

```
package com.cognizant.springlearn.util;  
  
import java.security.Key;  
import java.util.Date;  
  
import io.jsonwebtoken.Jwts;  
import io.jsonwebtoken.SignatureAlgorithm;  
import io.jsonwebtoken.security.Keys;  
  
public class JwtUtil {  
  
    private static final Key KEY =  
        Keys.secretKeyFor(SignatureAlgorithm.HS256);  
  
    public static String generateToken(String username) {  
  
        return Jwts.builder()  
            .setSubject(username)  
            .setIssuedAt(new Date())  
            .setExpiration(new Date(System.currentTimeMillis() + 600000))  
            .signWith(KEY)  
            .compact();  
  
    }  
}
```

AuthenticationController.java

```
package com.cognizant.springlearn.controller;  
  
import java.nio.charset.StandardCharsets;  
import java.util.Base64;  
  
import org.springframework.web.bind.annotation.GetMapping;  
import org.springframework.web.bind.annotation.RequestHeader;  
import org.springframework.web.bind.annotation.RestController;
```

```
import com.cognizant.springlearn.model.AuthenticationResponse;
import com.cognizant.springlearn.util.JwtUtil;
```

```
@RestController
```

```
public class AuthenticationController {
```

```
    @GetMapping("/authenticate")
```

```
    public AuthenticationResponse authenticate(
```

```
        @RequestHeader("Authorization") String authHeader) {
```

```
        String encoded = authHeader.substring(6);
```

```
        byte[] decodedBytes = Base64.getDecoder().decode(encoded);
```

```
        String credentials =
```

```
            new String(decodedBytes, StandardCharsets.UTF_8);
```

```
        String username = credentials.split(":")[0];
```

```
        String token = JwtUtil.generateToken(username);
```

```
        return new AuthenticationResponse(token);
```

```
    }
```

```
}
```

SecurityConfig.java

```
package com.cognizant.springlearn.config;
```

```
import org.springframework.context.annotation.Bean;
```

```
import org.springframework.context.annotation.Configuration;
```

```
import org.springframework.security.config.Customizer;
```

```
import org.springframework.security.config.annotation.web.builders.HttpSecurity;
```

```
import org.springframework.security.core.userdetails.User;
```

```
import org.springframework.security.core.userdetails.UserDetails;
```

```
import org.springframework.security.provisioning.InMemoryUserDetailsManager;
```

```
import org.springframework.security.web.SecurityFilterChain;
```

```
@Configuration
```

```
public class SecurityConfig {
```

```
    @Bean
```

```
    public InMemoryUserDetailsManager userDetailsService() {
```

```

        UserDetails user = User.withUsername("user")
            .password("{noop}pwd")
            .roles("USER")
            .build();

        return new InMemoryUserDetailsManager(user);
    }

    @Bean
    public SecurityFilterChain filterChain(HttpSecurity http)
        throws Exception {

        http
            .csrf(csrf -> csrf.disable())
            .authorizeHttpRequests(auth -> auth
                .requestMatchers("/authenticate").authenticated()
                .anyRequest().permitAll())
            .httpBasic(Customizer.withDefaults());

        return http.build();
    }
}

```

application.properties

```
server.port=8090
```

```
logging.level.org.springframework=INFO
```

```
logging.level.com.cognizant=DEBUG
```

SpringLearnApplication.java

```

package com.cognizant.springlearn;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class SpringLearnApplication {

    public static void main(String[] args) {

        SpringApplication.run(SpringLearnApplication.class, args);
    }
}

```


}

}